2021 Ph H2 Q6

Section: Particles and Waves

Topic: The Standard Model

This question involves identifying particles in the Standard Model.

(a)
State the name of a baryon

✅ A **baryon** is a particle made of **three quarks**.
Examples include:

- **Proton**
- **Neutron**

Answer: **Proton** (or **Neutron**)

(b)
State the name of a meson

✅ A **meson** is made of **a quark and an antiquark**.
Examples:

- **Pion**
- **Kaon**

Answer: **Pion** (or **Kaon**)

(c)
State the name of a fundamental particle that does not experience the strong nuclear force

✅ The **electron** is a fundamental particle (a lepton) that:

- Does **not** experience the **strong** nuclear force
- Is affected by electromagnetic and weak nuclear forces

Answer: **Electron**

🧠 Revision Tips

- **Baryons**: 3 quarks (e.g. proton, neutron)
- **Mesons**: 1 quark + 1 antiquark (e.g. pion, kaon)
- **Leptons** (e.g. electron, muon) are fundamental and **do not feel the strong force**
- In the Standard Model:
  - **Fermions** = matter particles (quarks & leptons)
  - **Bosons** = force carriers (e.g. photon, gluon)
